

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Expense Manager

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Expense Manager**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date:10 May,2026.....

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Expense Manager is a smart and intelligent web-based financial management system developed to help users efficiently track, manage, and analyze their daily expenses. In today's digital era, financial planning and expense tracking have become essential for students, working professionals, families, and small business owners. Traditional methods such as notebooks and spreadsheets are time-consuming, difficult to maintain, and often result in inaccurate budgeting and poor financial analysis. The Expense Manager system addresses these challenges by providing an automated and user-friendly platform for effective financial management.

The application is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. React.js is used for developing a responsive and interactive user interface, while Node.js and Express.js are used for backend development and API communication. MongoDB is used as a NoSQL database for securely storing user information, expense records, and transaction history.

The platform includes advanced features such as categorized expense tracking, real-time budget monitoring, graphical data visualization using pie charts and bar graphs, AI-powered chat support, secure authentication, search and filter functionality, CSV export, and large data import handling for scalability and ease purposes. Users can classify expenses into categories such as Food, Travel, Shopping, Bills, Health, Entertainment, and Others to better understand spending patterns and maintain budgeting discipline.

The AI-powered chat assistant helps users with financial guidance, application support, and expense-related queries. The application also supports responsive light/dark mode user interfaces for better accessibility across desktops, tablets, and mobile devices.

The main objective of the Expense Manager system is to improve financial awareness, simplify budgeting, reduce manual effort, and help users make better financial decisions through automation and analytics. By integrating modern technologies with secure data handling and intelligent support, Expense Manager provides a scalable, reliable, and efficient solution for personal and professional financial management.

1. INTRODUCTION

1.1 Overview

Expense Manager is a modern financial management application designed to simplify the process of tracking and managing daily expenses. Many individuals still rely on traditional methods such as notebooks, diaries, or spreadsheets for maintaining financial records. These methods are difficult to manage regularly and often lead to errors, overspending, and inaccurate financial planning.

The Expense Manager application provides a smart and automated solution that helps users maintain proper financial records and analyze spending habits effectively. The system allows users to securely register and log into the application using authentication mechanisms. After successful login, users can add, update, delete, and categorize expenses under categories such as Food, Travel, Shopping, Health, Entertainment, Bills, and Others.

1.2 Need of the Project

Managing expenses manually is time-consuming and less efficient. Most traditional methods lack automation, graphical analysis, and intelligent support systems. Users often fail to maintain proper records of expenses, which affects budgeting and financial planning.

The Expense Manager system is developed to overcome these limitations by providing:

- Automated expense tracking
- Real-time budget monitoring
- Graphical expense analysis
- Secure authentication and data storage
- AI-powered support system
- Efficient handling of larger financial imports

1.3 Objectives of the Project

The main objectives of the Expense Manager system are:

- To simplify expense tracking and financial management
- To provide categorized expense analysis
- To improve budgeting and financial awareness
- To reduce manual effort in maintaining records
- To provide graphical visualization of expenses
- To integrate AI-powered chat support
- To ensure secure data handling and authentication
- To support scalability and large import management

1.4 Problem Statement

1. Manual Expense Tracking

Traditional methods such as notebooks and spreadsheets are time-consuming, error-prone, and difficult to manage regularly.

2. Lack of Financial Insights

Most traditional systems do not provide graphical analysis, spending reports, or real-time budget monitoring.

3. Lack of Smart Automation

Many systems do not include AI-based assistance, intelligent alerts, or automated budgeting features.

4. Difficulty in Managing Large imports

Handling large amounts of expense records manually becomes complex and inefficient.

5. Security and Privacy Concerns

Users require secure authentication and protected storage of financial information.

2. METHODOLOGY

2.1 System Architecture

The Expense Manager application follows a client-server architecture developed using the MERN stack. The frontend communicates with the backend using RESTful APIs, ensuring smooth data transfer and efficient application performance.

2.2 Software Development Life Cycle (SDLC)

The project follows the Software Development Life Cycle (SDLC), which includes:

Requirement Analysis

User requirements and system functionalities were identified and analyzed.

System Design

Database structures, frontend layouts, APIs, and application workflow were planned.

Development

Frontend components, backend services, authentication systems, and database modules were implemented.

Testing

The application was tested for functionality, performance, and security.

Deployment and Maintenance

The final system was optimized and maintained for scalability and improved performance.

2.3 Working of the System

1. User registers and logs into the application.
2. Users add and manage expense records.
3. Expenses are categorized automatically.
4. Budget monitoring checks spending limits in real time.
5. Graphical charts analyze spending patterns.
6. AI-powered chat assistant provides support and guidance.
7. Expense reports can be searched, filtered, and exported.

3. TOOLS AND TECHNOLOGIES

3.1 Frontend Technologies

React.js

Used for building dynamic and responsive user interfaces.

Tailwind CSS

Used for modern and responsive UI design.

HTML5 and CSS3

Used for structuring and styling webpages.

JavaScript

Used for frontend functionality and interaction.

3.2 Backend Technologies

Node.js

Used for backend runtime and server-side operations.

Express.js

Used for API development and backend routing.

3.3 Database

MongoDB

Used for storing user information, expense records, and transaction history securely.

3.4 Development Tools

- Visual Studio Code
- Postman
- GitHub
- npm

3.5 Security and Authentication

- JWT Authentication
- Protected User Sessions
- Secure API Access

4. IMPLEMENTATION

4.1 User Authentication Module

The application provides secure signup and login functionality using JWT-based authentication. User credentials are securely managed to ensure privacy and protection of financial records.

4.2 Expense Management Module

Users can add, update, delete, and categorize expenses under multiple categories such as Food, Travel, Shopping, Bills, and Entertainment. The system stores transaction records efficiently and provides easy access to financial history.

4.3 Budget Monitoring System

The application monitors spending patterns and provides alerts whenever expenses exceed predefined budget limits. This helps users maintain financial discipline and control overspending.

4.4 Graphical Analytics

The dashboard provides graphical visualization through pie charts and bar graphs. These reports help users analyze monthly expenses, category-wise spending, and financial trends effectively.

4.5 AI Chat Integrated System

The AI-powered chat assistant helps users with financial queries, navigation support, and expense-related guidance. This intelligent support system improves overall user interaction and accessibility.

4.6 Large Import Handling

The application supports importing and processing larger import for less effort, scalability, and efficient financial data management. The system is optimized to maintain smooth performance while handling large volumes of records.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

- Successfully developed a smart and scalable expense management platform using the MERN stack.
- Improved financial tracking and budgeting efficiency.
- AI-powered chat support enhanced user interaction and assistance.
- Graphical analytics improved understanding of spending patterns and financial activities.
- Large import support improved scalability and application performance.
- Secure authentication ensured privacy and protection of financial records.

- Real-time budget monitoring helped users avoid overspending.
- Search, filter, and CSV export functionalities improved accessibility and financial reporting.
- Responsive UI improved usability across multiple devices.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

Expense Manager is a smart and intelligent financial management platform that simplifies expense tracking, budgeting, and financial analysis. By integrating MERN stack technologies, AI-powered support, graphical analytics, and secure authentication, the platform provides an efficient and user-friendly solution for financial management.

The project successfully demonstrates the implementation of modern web technologies for improving financial awareness and budgeting practices. The system reduces manual effort, improves financial discipline, and helps users make better financial decisions through automation and analytics

6.2 Future Scope

Future enhancements of the project may include:

- AI-based expense prediction and recommendations
- Mobile application development for Android and iOS
- Cloud deployment and scalability improvements
- Bank API integration for automatic transaction tracking
- Voice assistant and multilingual support
- Advanced analytics and personalized budgeting suggestions
- Notification and reminder system for bills and expenses

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